

Signal calibration from Pockel-cell

By Gage

Introduction

- SLUG sends signal into Pockels-cell modulator to a photodiode into the scope
- Modulators transmission is nonlinear (sin² curve)
- With Transmission governed by
- With Phase shift governed by

Transmission

$$T(V) = \sin^2 \left(\frac{\pi V}{2V_{\pi}} \right)$$

Phase Shift

$$\Delta\phi = \frac{2\pi}{\lambda} \Delta n(E) L$$

Equations

- First the voltage from the SLUG is time dependent

$$T(V) = \sin^2 \left(\frac{\pi V}{2V_{\pi}} \right)$$



$$T(t) = \sin^2 \left(\frac{\pi v_{\text{signal}}(t)}{2V_{\pi}} \right)$$

- Then because of bias with different V_{π}

- Where we can define the argument as

$$T(t) = \sin^2 \left[\frac{\pi}{2} \frac{V_{\text{bias}}}{V_{\pi, \text{bias}}} + \frac{\pi}{2} \frac{v_{\text{signal}}(t)}{V_{\pi, \text{RF}}} + \theta_0 \right]$$

$$\Theta(t) = \frac{\pi}{2} \frac{V_{\text{bias}}}{V_{\pi, \text{bias}}} + \frac{\pi}{2} \frac{v_{\text{signal}}(t)}{V_{\pi, \text{RF}}} + \theta_0$$

- Then we can invert the transmission

$$\sqrt{T} = |\sin \Theta|$$



$$\Theta(t) = \sin^{-1} \left(\sqrt{T(t)} \right)$$



$$\sin^{-1} \left(\sqrt{T(t)} \right) = \frac{\pi}{2} \frac{V_{\text{bias}}}{V_{\pi, \text{bias}}} + \frac{\pi}{2} \frac{v_{\text{signal}}(t)}{V_{\pi, \text{RF}}} + \theta_0$$

Or using trig identity

$$v_{\text{signal}}(t) = \frac{2V_{\pi, \text{RF}}}{\pi} \left[\sin^{-1} \left(\sqrt{T(t)} \right) - \frac{\pi}{2} \frac{V_{\text{bias}}}{V_{\pi, \text{bias}}} - \theta_0 \right]$$

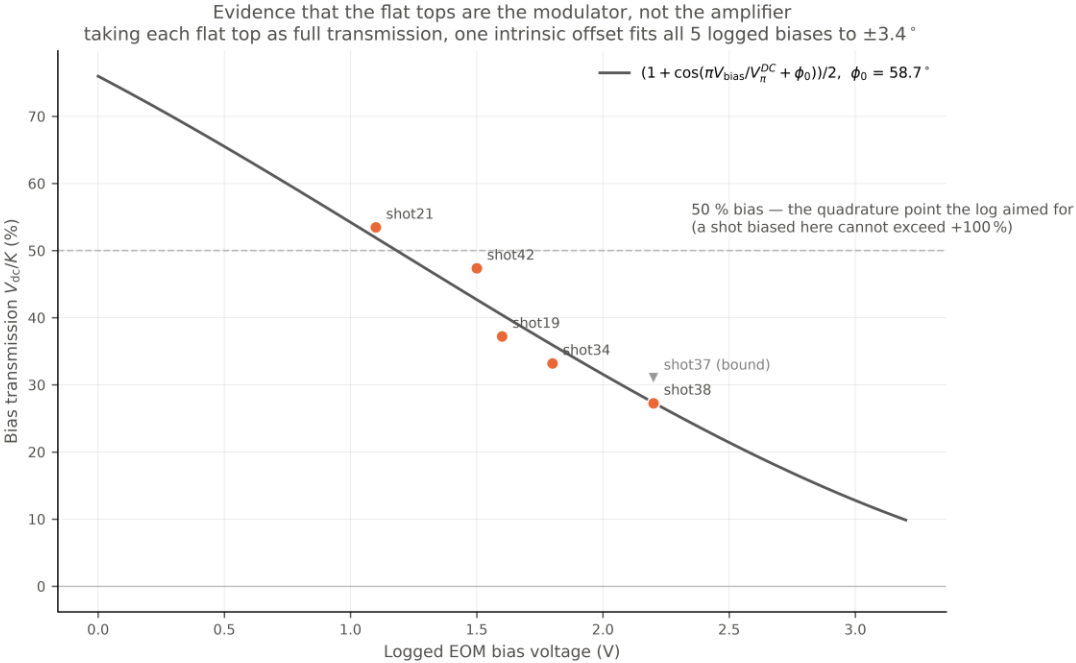
$$v_{\text{signal}}(t) = \frac{V_{\pi, \text{RF}}}{\pi} \left[\cos^{-1}(2T(t) - 1) - \pi \frac{V_{\text{bias}}}{V_{\pi, \text{bias}}} - \phi_0 \right]$$

Constants

Quantity	Value	How obtained
$V_{\pi,bias}$	6.14 V	Direct DC bias sweep
$V_{\pi,RF}$	~ 4.0 V	Working value / datasheet
ϕ_0	58.7° ?	Fit to EOM calibration data

Per Waveform

Quantity	How obtained
V_{bias}	Shot log / EOM bias setting
$V_{PD}(t)$	Recorded photodiode waveform
$V_{PD,baseline}$	Mean of pre-pulse region
V_{max}	Direct max-transmission measurement or inferred from baseline
$T(t)$	Calculated from $V_{PD}(t)/V_{max}$



$$T_{bias} = \frac{1 + \cos\left(\pi \frac{V_{bias}}{V_{\pi,bias}} + \phi_0\right)}{2}$$

$$T(t) = \frac{V_{PD}(t)}{V_{max}}$$

$$V_{max} = \frac{V_{PD,baseline}}{T_{bias}}$$

Implementation

1. Load the waveform
2. Normalize to optical transmission
3. Invert the transfer function
4. Recover original signal
5. Plug corrected waveforms into working directory convert to energy spectrum

$$T(t) = \frac{V_{\text{PD}}(t)}{V_{\text{max}}}$$

$$\phi(t) = \cos^{-1} (2T(t) - 1)$$

$$v_{\text{signal}}(t) = \frac{V_{\pi, \text{RF}}}{\pi} [\phi(t) - \phi_{\text{bias}}]$$

$$\phi_{\text{bias}} = \pi \frac{V_{\text{bias}}}{V_{\pi, \text{bias}}} + \phi_0$$

Preliminary results (shot 9)

- Still working through some kinks
- Transmission was greater than 100 percent in unstable region (Saturation?)

Shot 9 -- raw vs. corrected (LNX1020F: bias=6.14V, RF=4.0V, phi0=58.7deg)

